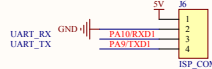
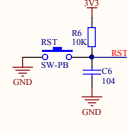


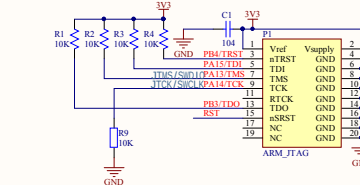
供电稳压供电



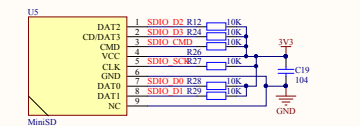
ISP串口下载程序接口



复位按键



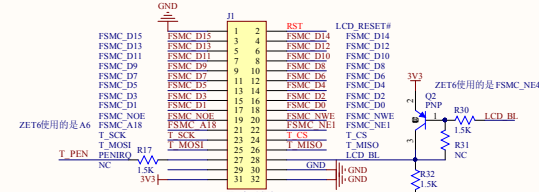
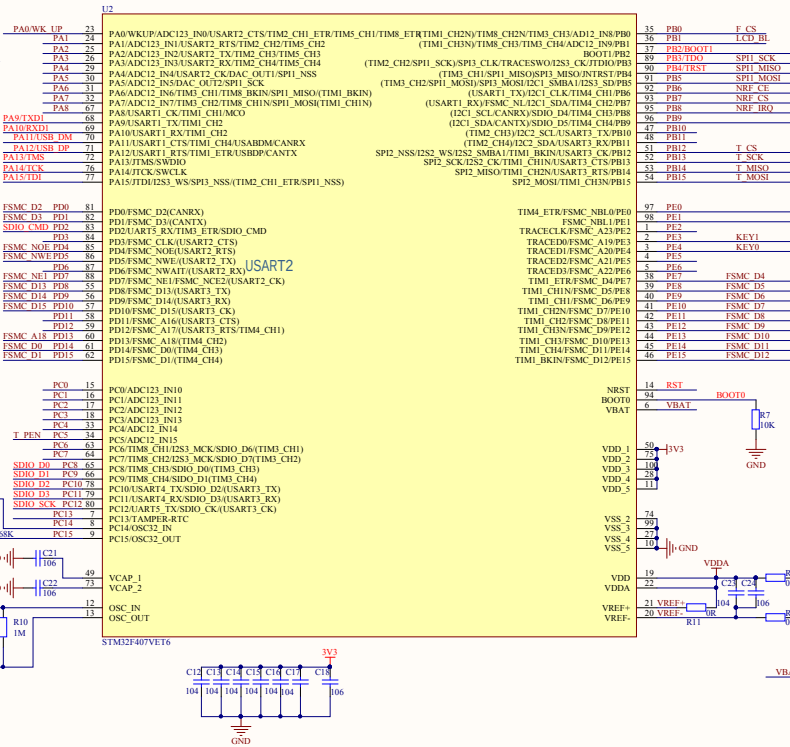
JTAG/SWD调试接口



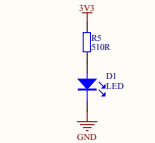
预留IO接口

## CPU简介

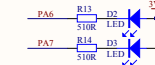
STM32F407VET6  
RAM: 196K FLASH: 512K  
内核: ARM 32位的 Cortex-M3 CPU  
最高168MHz工作频率



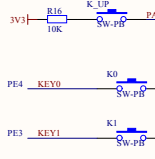
TFT液晶屏接口



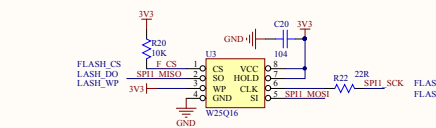
供电指示灯



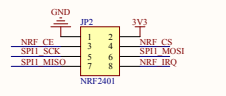
LED指示灯



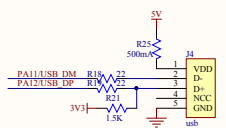
用户按键



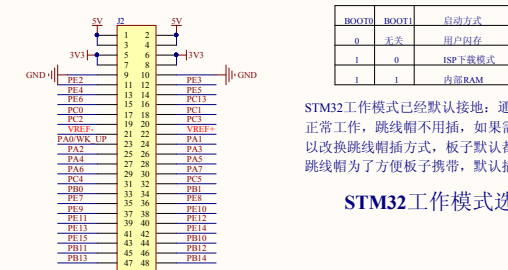
外部存储FLASH



NRF2401无线模块接口

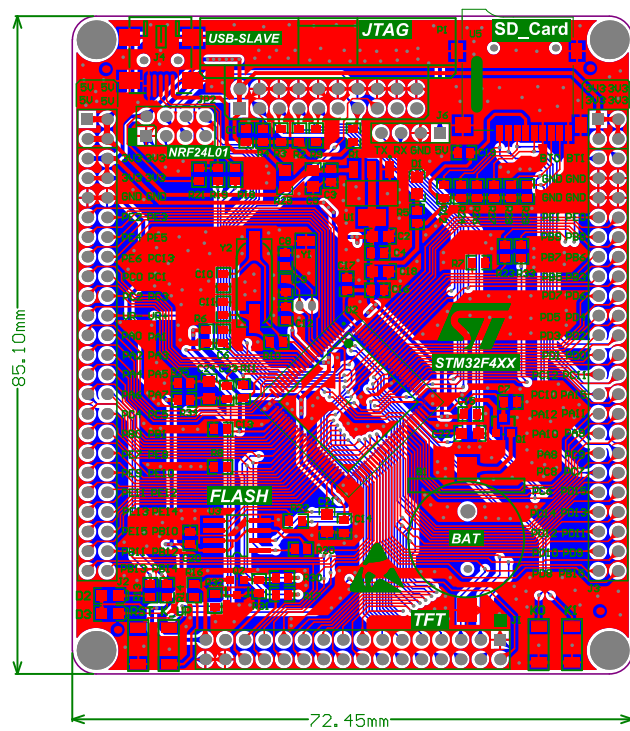


STM32-USB Slave接口



预留IO接口

| Title | Size                                       | Number   | Revision |
|-------|--------------------------------------------|----------|----------|
| A2    | 2016-1-5                                   | 1        |          |
| File: | C:\Documents and Settings\Administrator\My | Sheet of |          |



| Comment       | Description                                                                                                                                               | Designator                                                             | Footprint     | LibRef                                                                                          | Quantity |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------|----------|
| BAT           |                                                                                                                                                           | B1                                                                     | BAT           | BATTERY                                                                                         | 1        |
| 104           | Capacitor, Capacitor, Capacitor, Capacitor, Capacitor, Capacitor, Capacitor, Capacitor, Capacitor (Semiconductor SIM Model), [NoValue], Capacitor         | C1, C3, C4, C6, C7, C12, C13, C14, C15, C16, C17, C19, C20, C23        | RC0603        | Cap, CAP, CAP, CAP, Cap, Cap, Cap, Cap, Cap, Cap, Cap, Cap Semi, CAP, Cap                       | 14       |
| 106           | [NoValue], [NoValue], Capacitor, Capacitor, Capacitor, Capacitor                                                                                          | C2, C5, C18, C21, C22, C24                                             | RC0805        | ELECTRO2, ELECTRO2, Cap, Cap, Cap, Cap                                                          | 6        |
| 12P           |                                                                                                                                                           | C8, C9                                                                 | RC0603        | CAP                                                                                             | 2        |
| 22P           | Capacitor                                                                                                                                                 | C10, C11                                                               | RC0603        | CAP                                                                                             | 2        |
| LED           |                                                                                                                                                           | D1, D2, D3                                                             | LED0805       | LED                                                                                             | 3        |
| Header 16X2   | Header, 16-Pin, Dual row                                                                                                                                  | J1                                                                     | IDE2X16       | Header 16X2                                                                                     | 1        |
| Header 24X2   | Header, 24-Pin, Dual row                                                                                                                                  | J2, J3                                                                 | IDE2X24       | Header 24X2                                                                                     | 2        |
| usb           |                                                                                                                                                           | J4                                                                     | Mini_USB      | usb                                                                                             | 1        |
| ISP_COM       | Header, 4-Pin                                                                                                                                             | J6                                                                     | HDR1X4        | Header 4                                                                                        | 1        |
| NRF2401       | Header, 4-Pin, Dual row                                                                                                                                   | JP2                                                                    | IDE2X4        | Header 4X2                                                                                      | 1        |
| SW-PB         |                                                                                                                                                           | K0, K1, K_UP, RST                                                      | SW-SPST       | SW-PB                                                                                           | 4        |
| ARM_JTAG      | JTAG                                                                                                                                                      | P1                                                                     | JTAG-20P      | JTAG                                                                                            | 1        |
| BAT54C        |                                                                                                                                                           | Q1                                                                     | SOT-23-BAT54C | BAT54C                                                                                          | 1        |
| PNP           | PNP Bipolar Transistor                                                                                                                                    | Q2                                                                     | PNP_SOT23     | PNP                                                                                             | 1        |
| 10K           | [NoValue], [NoValue], [NoValue], [NoValue], Resistor, Resistor, Resistor, Resistor, Resistor, Resistor, [NoValue], Resistor, Resistor, Resistor, Resistor | R1, R2, R3, R4, R6, R7, R8, R9, R12, R16, R20, R24, R26, R27, R28, R29 | RC0603        | RES, RES, RES, RES, Res2, Res2, Res2, Res2, Res2, Res2, RES, Res2, Res2, Res2, Res2, Res2, Res2 | 16       |
| 510R          | Resistor                                                                                                                                                  | R5, R13, R14, R23, R33                                                 | RC0603        | Res2                                                                                            | 5        |
| 1M            | Resistor                                                                                                                                                  | R10                                                                    | RC0603        | Res2                                                                                            | 1        |
| 0R            | Resistor                                                                                                                                                  | R11, R34, R35                                                          | RC0603        | Res2                                                                                            | 3        |
| 1.5K          |                                                                                                                                                           | R17, R21, R30, R32                                                     | RC0603        | RES                                                                                             | 4        |
| 22            |                                                                                                                                                           | R18, R19                                                               | RC0603        | RES                                                                                             | 2        |
| 22R           |                                                                                                                                                           | R22                                                                    | RC0603        | RES                                                                                             | 1        |
| 500mA         |                                                                                                                                                           | R25                                                                    | RC0805        | RES                                                                                             | 1        |
| NC            |                                                                                                                                                           | R31                                                                    | RC0603        | RES                                                                                             | 1        |
| AMS1117-3.3V  |                                                                                                                                                           | U1                                                                     | SOT-223       | 11                                                                                              | 1        |
| STM32F407VET6 |                                                                                                                                                           | U2                                                                     |               | STM32F407VET6                                                                                   | 1        |
| W25Q16        |                                                                                                                                                           | U3                                                                     | SOP8L         | W25X16                                                                                          | 1        |
| MiniSD        |                                                                                                                                                           | U5                                                                     | TF_SD         | MiniSD                                                                                          | 1        |
| 32.768K       | 晶振                                                                                                                                                        | Y1                                                                     | JZ32768       | CRYSTAL                                                                                         | 1        |
| 8M            | Crystal                                                                                                                                                   | Y2                                                                     | 49S_SMD       | CRYSTAL                                                                                         | 1        |